DY PATIL
DEEMED TO BE
UNIVERSITY
NAVI MUMBAI

**Semester - III** | **B. Tech. Computer Engineering** | **39724303**

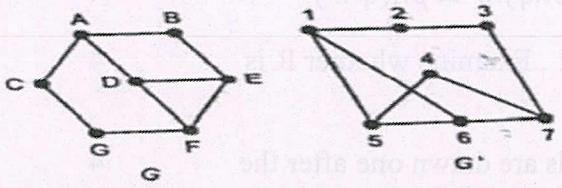
**2413CEC3T3** | **Discrete Mathematics**

**Date: 06/06/2026** | **Time: 02:00 PM TO 05:00 PM** | **Marks:100**

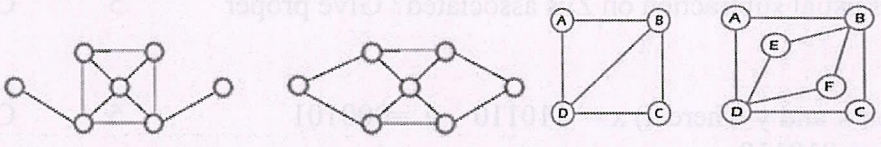
| | **Q.1 Attempt all questions** | **20 Marks** | **CO** | **BT** |
|---|---|---|---|---|
| **a)** | If p, q and r are propositions , then show that $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ | 4 | CO1 | BT3 |
| **b)** | A relation R in Z is defined by x R y iff x < y+1 . Examine whether R is reflexive, symmetric and transitive? | 4 | CO2 | BT3 |
| **c)** | From a well shuffled pack of 52 cards, two cards are drawn one after the another without replacement. Find the probability that i) Both are king cards ii) the first is a king card and the second is a queen. | 4 | CO3 | BT3 |
| **d)** | Define the following with example<br>i) Planar graph ii) Hamiltonian graph | 4 | CO4 | BT2 |
| **e)** | Find the weights of the words given below :<br>i) 11111 ii) 010101 ii) 0110 | 4 | CO6 | BT3 |

| | **Q.2 Attempt Any 4** | **20 Marks** | **CO** | **BT** |
|---|---|---|---|---|
| **a)** | Prove the logical equivalence $(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \equiv r$ | 5 | CO1 | BT3 |
| **b)** | Define a relation R on the set Z by aRb if a-b is a non-negative even integer. Verify whether R is a partial order relation. | 5 | CO2 | BT4 |
| **c)** | The probabilities that students A, B, C, D solve a problem are $\frac{1}{3}, \frac{2}{5}, \frac{1}{4}$ and $\frac{1}{5}$ respectively. If all of them try to solve the problem, what is the probability that the problem is solved? | 5 | CO3 | BT3 |
| **)** | Prove that the set of cube roots of unity is a group under multiplication of complex numbers. | 5 | CO4 | BT3 |
| **e)** | Is the binary operation of usual subtraction on Z is associated? Give proper reason with example. | 5 | CO5 | BT3 |
| **f)** | Find the distance between x and y where i) x = 110110 , y =000101<br>ii) X=001100 , y =010110 | 5 | CO6 | BT4 |

| | **Q.3. Attempt Any 4** | **20 Marks** | **CO** | **BT** |
|---|---|---|---|---|
| **a)** | Prove by mathematical induction<br>$1^3 + 2^3 + \cdots \ldots \ldots n^3 = \frac{n^2(n+1)^2}{4}$ | 5 | CO1 | BT3 |
| **b)** | Draw the Hasse diagram of the poset A = {2,3,6,12,24,36,72} under the relation of divisibility. Is it a lattice ? | 5 | CO2 | BT3 |

| | | | | |
|---|---|---|---|---|
| **c)** | The chances that A, B, and C will be the Education Minister of Government of India are in the ratio 4 : 1 : 2. The probabilities that they will introduce reservations in professional colleges for backward classes are 0.3, 0.8 and 0.5 respectively. Find the probability that the bill for reservation will be introduced? | 5 | CO3 | BT3 |
| **d)** | Check and explain the following graphs are isomorphic or not   G G' | 5 | CO4 | BT3 |
| **e)** | Prove that the set A = { 0,1,2,3,4,5} is a finite Abelian group under addition modulo 6. | 5 | CO5 | BT3 |
| **f)** | Consider the following (2, 6) encoding function e. e(00) = 000000, e(10) = 101010 , e(01) = 011110 , e(11) = 111000 i) Find the minimum distance of e . ii) How many errors will e detect? | 5 | CO6 | BT3 |

| **Q.4.** | **Attempt Any 2** | **20 Marks** | **CO** | **BT** |
|---|---|---|---|---|
| **a)** | Verify the validity of the argument : If you invest in the stock market, then you will get rich. If you get rich, then you will be happy. ∴ If you invest in stock market, then you will be happy. | 10 | CO1 | BT4 |
| **b)** | Functions f and g are defined as follows: $f: R \to R \ , g \ : R \to R \ , f(x) = 2x + 3, g(x) = 3x - 4$ Find $fog, f^{-1}, g^{-1}$ *and verify that* $(fog)^{-1} = g^{-1}of^{-1}$ | 10 | CO2 | BT4 |
| **c)** | A bag contains 7 red and 3 black balls and another bag contains 4 red and 5 black balls. One ball is transferred from the first bag to the second bag and then a ball is drawn from the second bag. If this ball happens to be red, find the probability that a black ball is transferred. | 10 | CO3 | BT3 |

| **Q.5.** | **Attempt Any 2** | **20 Marks** | **CO** | **BT** |
|---|---|---|---|---|
| **a)** |   Which of the above four graphs is Eulerian, Hamiltonian or both? | 10 | CO4 | BT3 |
| **b)** | Consider the set A = { 1, 2, 3, 4, 5, 6} under the multiplication modulo 7. I) Construct the multiplication table ii) Find the inverses of 2, 3, 5 and 6. Prove that A is a cyclic group. | 10 | CO5 | BT3 |
| **c)** | Show that the encoding function $e: B^2 \to B^4$ is a group code e(00) = 0000, e(10) = 0110, e(01) = 1011 , e(11) = 1100 | 10 | CO6 | BT3 |